

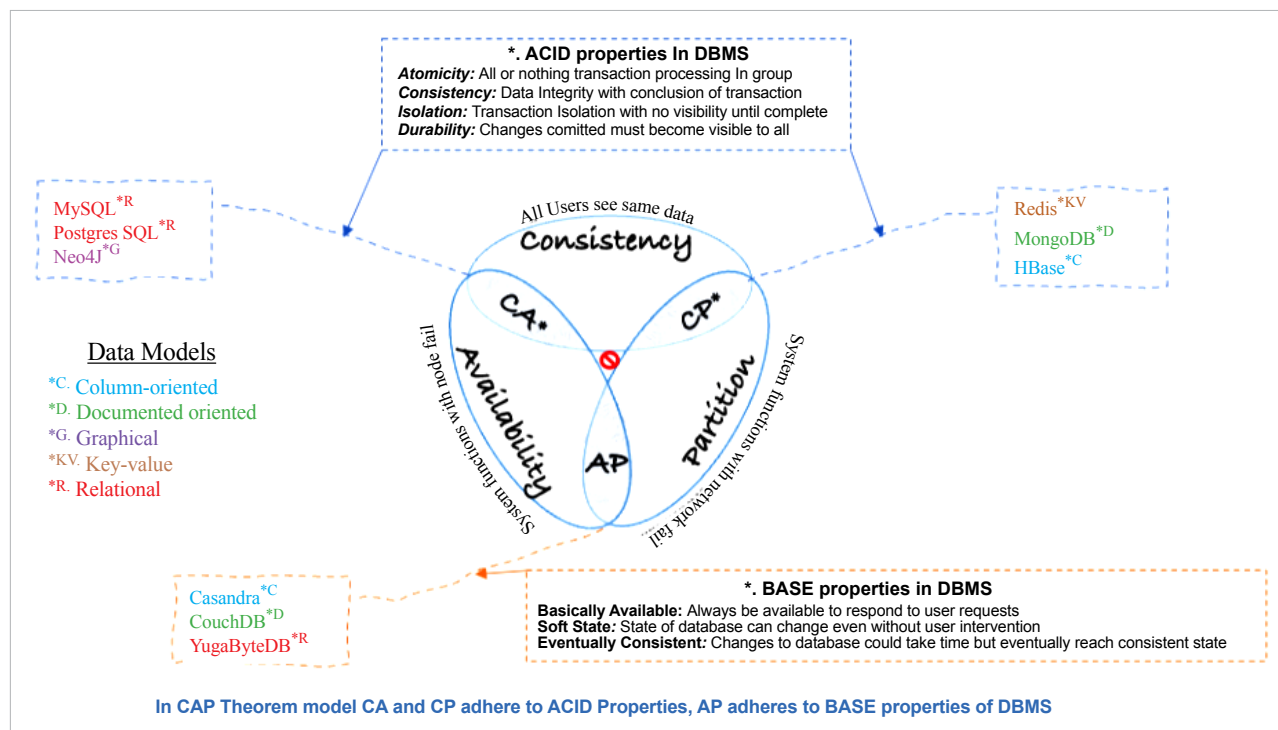


Figure 1: Distributed open source databases and CAP theorem overview

of various open source databases. It compares them in terms of data model, transaction model and adherence to the CAP theorem. While not comprehensive, this table provides context for comparing different databases against the use cases of interest for any distributed system.

It is important to note that while the CAP theorem offers valuable insights into the trade-offs in distributed systems, not every database strictly adheres to one of the three properties. Many databases offer different levels of consistency and availability depending on the specific requirements of the application.

Engineers and architects designing systems must have a thorough understanding of database properties to fulfil their basic functions effectively. Factors such as heavy reads vs light writes, heavy reads and heavy writes, high volume transactions, changes expected to schema, and more, define the basic characteristics of database requirements. Applying the

CAP theorem, ACID, BASE along with considering the data model and transaction model, can help identify the right set of databases that fit the needs. Balancing the trade-offs

between consistency, availability and partition tolerance while considering the proof-of-concept will ensure a solid foundation for a well-designed system that meets organisational needs. **END** 🐧

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